

9. (a) A student tested compounds **A**, **B** and **C** to identify them.

Her observations are recorded in the tables below.

Compound A	Observation	Ion present
Flame test	apple-green flame	barium
Add silver nitrate solution	white precipitate	chloride

Compound B	Observation	Ion present
Flame test	lilac flame	
Add silver nitrate solution	cream precipitate	

Compound C	Observation	Ion present
Flame test	brick-red flame	
Add silver nitrate solution	yellow precipitate	

- (i) Use your knowledge of the tests for ions to complete the tables for compounds **B** and **C**.

[3]

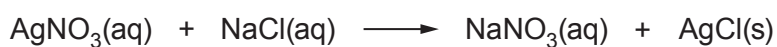
- (ii) Write the chemical formula for compound **A**.

[1]

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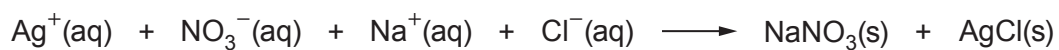
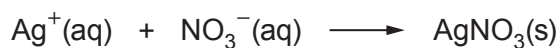
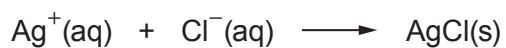
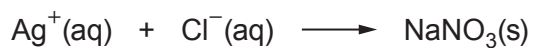


- (b) When silver nitrate solution reacts with sodium chloride solution to produce a white precipitate the following reaction occurs.



Put a tick (✓) in the box which shows the ionic equation for the formation of the precipitate.

[1]

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- (c) When silver carbonate is heated it breaks down to give silver metal and carbon dioxide and oxygen gas.



Calculate the mass of silver that would be produced from heating 13.8 g of silver carbonate.

[3]

$$M_r(\text{Ag}_2\text{CO}_3) = 276 \quad A_r(\text{Ag}) = 108$$

Mass = g

END OF PAPER

